

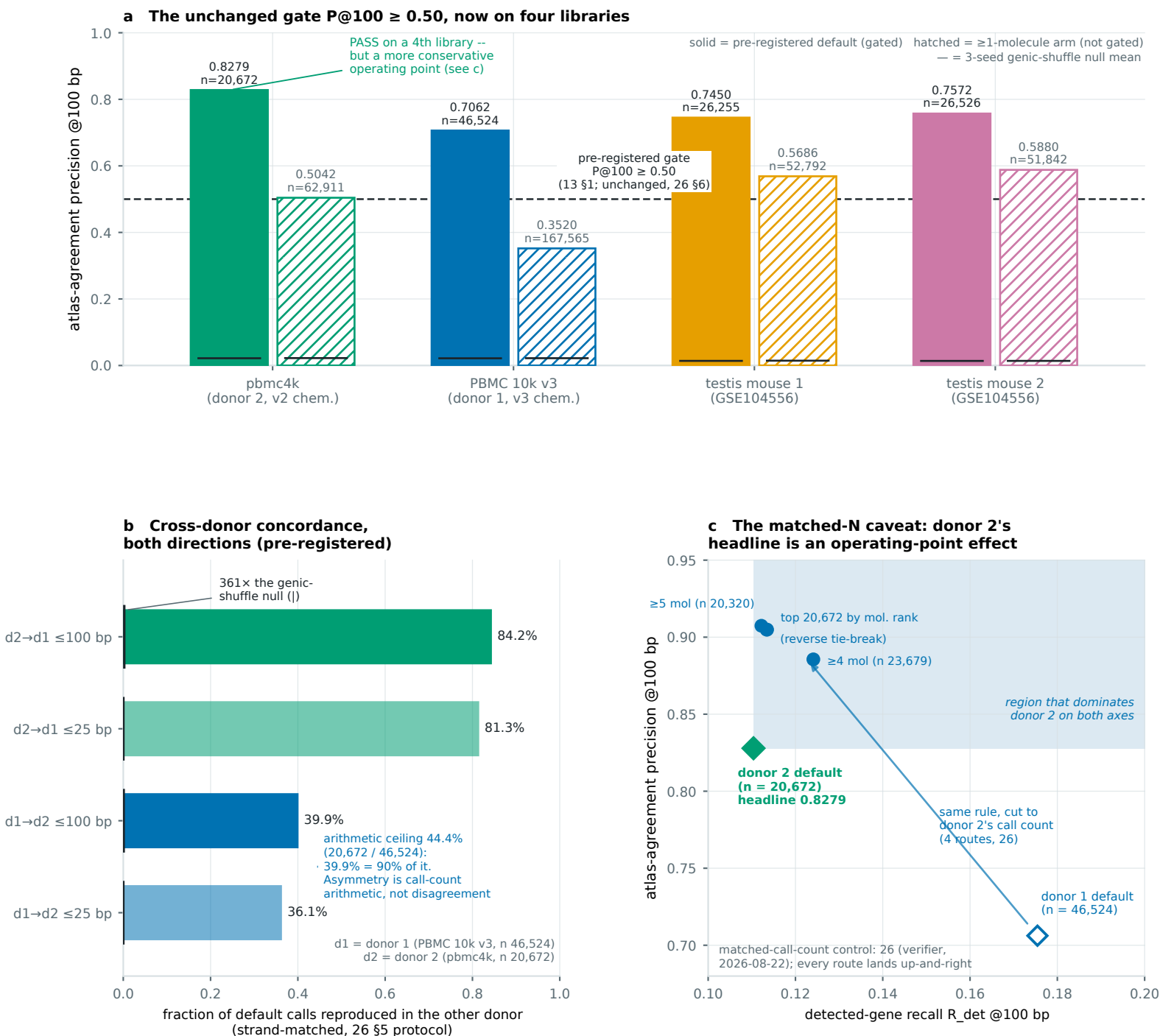


Fig S4 | Second-donor validation (pre-registered in 26 on 2026-08-21, BEFORE any pbmc4k number existed; run 2026-08-22; verifier verdict FIXED). (a) The 10x public pbmc4k library (v2 chemistry, Cell Ranger 2.1.0, 4,340 cells) scored by the UNCHANGED pre-registered precision default under the identical frozen code (9dfdefb3), scorer, atlas and nulls: 107 of 108 recorded parameters identical, the one change being --seq-len 91→98, a library descriptor. Gate $P@100 \geq 0.50$: PASS at 0.8279 (38× its null) — the gate now holds on four libraries. The ≥ 1 -molecule arm (hatched) again has the higher $F1_{\text{det}}$ (0.2635 vs 0.1948): ≥ 2 molecules is a reliability choice, not the $F1$ optimum. (b) Cross-donor concordance per the pre-registered \$5 protocol, both directions, never one alone: 84.2% of donor-2 default calls have a strand-matched donor-1 call within 100 bp (81.3% within 25 bp), ~361× the genic-shuffle null; the reverse direction is 39.9% (36.1% at 25 bp) against its own null and an arithmetic ceiling of 44.4% — donor 1 emits 46,524 default calls to donor 2's 20,672, so the asymmetry is call-count arithmetic, not disagreement. (c) The verifier's matched-N control, drawn so it cannot be buried: cut donor 1's default to donor 2's call count by ANY route (its own clip-molecule rank under both tie-breaks, ≥ 4 or ≥ 5 molecules) and donor 1 lands at $P@100$ 0.886–0.907 with R_{det} 0.112–0.124 — dominating donor 2 on both axes. Donor 2's higher headline precision is therefore an operating-point effect: the fixed rule lands at a more conservative point on the shallower v2 library. Precision GENERALISES; it does not improve. Scope (26): a second library, chemistry and Cell Ranger version — only presumptively a second individual (10x publishes no donor id). Recall is genuinely lower on donor 2 (R_{det} 0.1104 vs 0.1754; survives every denominator swap). Values: results/figures/manuscript/figS4_seconddonor*.tsv.